# CS316 Chapter 4 Start: First vs Second Order

The idea in simple words

## First order: the SLOPE (which way is downhill?)

- **f'(x) = slope**: how steep the curve is at a point. Positive = going uphill; negative = going downhill.
- **Gradient** (the upside-down triangle f): the same idea with several variables (e.g. w and b). A list of slopes, one per variable. It points in the direction that goes **uphill fastest**.
- **Jacobian**: first derivatives of a function with **several outputs**.

## Second order: the SHAPE (are we in a bowl?)

- **f''(x) = curvature**: does the curve bend up or down?
- **Hessian**: the same idea with several variables. A table (matrix) of second derivatives.
- **f'' > 0**: bends up like a bowl. The flat point (f' = 0) is the **bottom: a minimum**.
- **f'' < 0**: bends down like a hill. The flat point is the **top: a maximum**.
- **f'' = 0**: no bend at that point. You can still compute it (it is just 0), but the test is **inconclusive**: a line (no bottom), $x^3$ at 0 (not a bottom), x■ at 0 (a bottom).
- **Hessian PSD** (positive semidefinite) = bends up or stays flat in **every direction**. If true **everywhere**: the whole thing is one bowl (convex), so any bottom is the **global minimum**.

## Convex vs concave

| Shape | Picture | Flat point is | Note |
|---|---|---|---|
| Convex | Bowl (U) | Minimum | If a minimum exists it is the global one (e.g. MSE). Convex does not always guarantee a minimum exists (e^x) |
| Concave | Hill (upside-down U) | Maximum | The top is the global maximum |

## The classic test

| Step | Check | Meaning |
|---|---|---|
| 1 | f'(x) = 0 | A flat point: candidate min or max |
| 2 | f''(x) > 0 | Minimum (bends up) |
| 2 | f''(x) < 0 | Maximum (bends down) |
| 2 | f''(x) = 0 | Inconclusive |

## Gradient descent vs Newton

```
x_new = x_old - alpha * gradient      # gradient descent (FIRST order: slope only)
x_new = x_old - H^-1 * gradient       # Newton (SECOND order: slope + Hessian)
```

| | Gradient descent | Newton's method |
|---|---|---|
| Uses | Slope only (first order) | Slope + curvature (Hessian, second order) |
| Steps | Many cheap steps | Few expensive steps (compute and invert the Hessian) |
| Speed | Slower, steady | Faster near the minimum |
| Many variables | Scales well (gradient = n numbers) | Impractical (Hessian = n x n numbers) |
| Tuning | You choose the learning rate alpha | Step size set by the Hessian |
| Risk | alpha too big diverges; too small is slow | Can fail if the Hessian is not well behaved (non-convex) |

**In one line:** first order tells you which way is downhill; second order tells you whether you are in a bowl. Gradient descent uses only the first; Newton uses both. Gradient descent stops where the gradient = 0; for a convex loss like MSE that point is the global minimum.